

Phase space mixing in the presence of self-gravity: spherically symmetric Vlasov-Poisson evolutions with fixed and dispersed angular momentum

Erik Jiménez-Vázquez¹ and Olivier Sarbach²

¹ *[(Erik) Affiliation to be completed.]*

²*Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo, Edificio C-3, Ciudad Universitaria, 58040 Morelia, Michoacán, México*

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[(Erik) Provisional; to be rewritten when sections VI and VII are final.] This article is devoted to the numerical study of a spherically symmetric, collisionless kinetic gas which is trapped in an external central potential and whose self-gravity is taken into account. In the absence of self-gravity, such a gas relaxes in time to a stationary state through phase space mixing. We analyze how this process is modified by the self-gravity of the gas, considering both the reduced model in which all gas particles have the same magnitude of the angular momentum and the case in which the angular momenta are distributed over an interval. Our study is based on a particle-in-cell code, on a numerical solution of the linearized equations in action-angle variables, and on a spectral criterion which characterizes the existence of oscillating modes. We find that small perturbations of a self-consistent steady state are damped as long as the mass of the gas lies below a threshold, whereas above this threshold they give rise to an oscillation which is not damped, and we show that the threshold and the frequency of the oscillation are accurately predicted by the spectral criterion. Finally, we show that a dispersion in the angular momentum suppresses the oscillations which are present in the reduced model.

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I. INTRODUCTION

Collisionless kinetic gases provide the standard description for systems consisting of a large number of particles which interact with each other only through the mean field they generate. Prominent examples are the stars in a galaxy or in a globular cluster, and the dark matter particles in a galactic halo [1]. The state of such a gas at time t is described by its one-particle distribution function (DF) $f(t, x, p)$, a nonnegative function on the one-particle phase space which determines the number of gas particles in a small phase space volume about the point (x, p) , and whose evolution is governed by the Vlasov equation.¹ In the Newtonian gravitational case, the Vlasov equation is coupled to Poisson's equation for the gravitational potential generated by the gas, which yields the Vlasov-Poisson system, see Ref. [2] for a review of its mathematical properties. Its general relativistic counterpart is the Vlasov-Einstein system [3].

It is instructive to compare the dynamics of a self-gravitating collisionless gas with the one of a pressureless fluid (dust), which is obtained in the limit in which the velocity dispersion of the gas vanishes. The gravitational collapse of a spherical dust cloud leads to the formation of a singularity in finite time at which the density diverges, both in the Newtonian theory and in general relativity [4]; in the latter case, part of this singularity may even be visible to distant observers [5, 6]. In contrast to this, the velocity dispersion of a kinetic gas prevents the particles from focusing. In fact, the Vlasov-Poisson system admits global classical solutions for general initial data [7–9], while for the spherically symmetric Vlasov-Einstein system global solutions exist for small initial data [10] and sufficiently concentrated data collapse to a black hole [11], the threshold between these two outcomes having been explored numerically in Refs. [12–15]. Therefore, for a Newtonian collisionless gas the relevant question is not whether or not a singularity forms, but rather how the gas behaves for large times and, in particular, whether or not it relaxes to a stationary state.

A fundamental mechanism leading to such a relaxation is phase space mixing [16, 17]. To explain it, consider first a gas whose self-gravity is neglected and which is trapped in the potential well of an external potential. If the one-particle Hamiltonian is integrable, each gas particle is confined to an invariant torus, and in terms of action-angle variables (Q, J) the solution of the Vlasov equation is $f(t, Q, J) = F(Q - \Omega(J)t, J)$, where F describes the initial DF and $\Omega(J)$ denotes the frequencies associated with the torus labeled by J . Since these frequencies vary from one torus to the other, the DF is stretched along the angle directions and develops structures on finer and finer scales as time increases. As a consequence, f does not converge pointwise as $t \rightarrow \infty$. However, the macroscopic observables which are obtained by integrating f against a test function do converge, and their limit is the one associated with

¹ The Vlasov equation is also often called the collisionless Boltzmann equation in the literature.

the average of the initial DF over the angle variables, which depends only on integrals of motion. A mathematically precise formulation of this statement and its proof for rotationally symmetric potentials in $d = 1, 2, 3$ space dimensions satisfying a suitable non-degeneracy condition were given by Rioseco and Sarbach [18] (see also Refs. [19, 20]). The results in Ref. [18] include the case of a spherically symmetric gas configuration in which all the gas particles have the same value of the total angular momentum, which behaves like an effective one-dimensional system, and they explain the relaxation to a virialized state which had been observed in the numerical simulations of Ref. [21]. For the analogous phenomenon in the exterior of a black hole we refer the reader to Refs. [22, 23], and for estimates on the rate at which the observables converge to Refs. [24–26].

The generalization to the self-gravitating case, which was left open in Ref. [18], is considerably more involved, since the potential now depends on the DF, such that a perturbation of the latter induces a perturbation of the potential which, in turn, acts back on the gas. In plasma physics, where the particles interact through their mean electric field, the corresponding collective effect has been known since the work of Landau [27]: for a gas of charged particles close to a spatially homogeneous equilibrium the electric field decays in time while the DF undergoes mixing. This phenomenon, known as Landau damping, has been established rigorously at the nonlinear level by Mouhot and Villani [28]. The gravitational case differs from the electromagnetic one in two important aspects. First, the interaction is attractive, such that the collective response of the gas tends to reinforce a perturbation instead of screening it. Second, a self-gravitating gas configuration of finite mass does not possess spatially homogeneous equilibria. Instead, its steady states are spatially inhomogeneous, the gas particles are trapped in the potential well, and the frequencies of their radial motion lie in an interval which, in the situations considered in this article, is bounded and bounded away from zero. For this reason, a collective oscillation whose frequency lies below this interval is not in resonance with any gas particle and hence cannot be damped by mixing.

The question of whether small perturbations of a steady state of the gravitational Vlasov-Poisson system are damped or give rise to persistent oscillations goes back to Lynden-Bell [16] and has received considerable attention in recent years. Hadžić, Rein and Straub [29] (see also Refs. [30, 31]) showed that the essential spectrum of the operator governing the linearized dynamics about a large class of spherically symmetric steady states is determined by the radial periods of the gas particles. Using a version of the Birman-Schwinger principle which was first applied to this problem by Mathur [32], they derived a criterion for the existence of an eigenvalue in the gap below this spectrum, which corresponds to an oscillating mode that is not damped. This criterion is formulated in terms of a family of bounded operators acting on functions of the radius. Related criteria for the existence of oscillating modes and estimates for their number, which also allow for an attractive external potential, have been given by Moreno, Rioseco and Van Den Bosch [33] for steady states whose DF depends only on the energy. For the reduced system in which all the gas particles have the same magnitude of the angular momentum and move around a central point mass, Hadžić, Rein, Schrecker and Straub [34] proved the following dichotomy for steady states of sufficiently small mass whose DF is proportional to $(E_0 - E)^k$ for energies E below the cutoff energy E_0 : the solutions of the linearized system are damped if $k > 1$, whereas an oscillating mode exists if $1/2 < k \leq 1$. Quantitative decay rates for the linearized system about steady states with a central point mass and a distribution of angular momenta have been obtained in Ref. [35], and for the nonlinear system with an external Kepler potential, phase mixing of spherically symmetric data over long time scales has been established in Ref. [36]. On the numerical side, Ramming and Rein [37] and Straub [38] have performed particle-in-cell simulations of the spherically symmetric Vlasov-Poisson system which show that, depending on the steady state, its perturbations either oscillate without damping or decay. Finally, we mention that weakly damped non-radial modes of spherical stellar systems have been computed in Refs. [39, 40] by means of the matrix method.

In this article, we study numerically the transition between mixing and persistent oscillations for a spherically symmetric collisionless gas which is trapped in an external central potential and whose self-gravity is taken into account. We consider two situations. The first one is the reduced model in which all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum, which has a two-dimensional phase space and is the setting considered in Refs. [18, 34]. In the second situation the angular momenta of the gas particles are distributed over an interval. For the external potential we choose the isochrone potential, for which the action-angle variables are known explicitly, and the potential generated by a point mass. Our analysis is based on three independent tools: (i) a particle-in-cell (PIC) code which solves the nonlinear Vlasov-Poisson system, (ii) a numerical solution of the equations linearized about a self-consistent steady state, formulated in the action-angle variables of the latter, and (iii) the numerical computation of the largest eigenvalue $\lambda(\omega)$ of the operator of Mathur, which measures the gain of the self-consistent loop in which a perturbation of the potential with frequency ω induces a perturbation of the density which, in turn, generates a potential. An oscillating mode with frequency below the frequency band of the gas particles exists if and only if the value λ_{edge} of this gain at the lower edge of the band is larger than one.

The main results of this article can be summarized as follows. First, as long as the mass of the gas is sufficiently small compared to the mass which generates the external potential, a small perturbation of a self-consistent steady state is damped, and the frequency and the damping rate of the collective response obtained from the PIC simulations

agree with those obtained from the linearized equations to within one percent. Second, for the reduced model there is a threshold mass above which a discrete mode emerges below the frequency band, such that the perturbation oscillates without being damped. We show that this threshold is characterized by the condition $\lambda_{edge} = 1$, and that the frequency of the mode predicted by the equation $\lambda(\omega) = 1$ agrees with the one measured in the simulations. Third, for steady states of small mass with $k \leq 1$, the oscillating mode whose existence is guaranteed by the results in Ref. [34] lies so close to the edge of the band that it cannot be distinguished from the decaying part of the response over the time span of our simulations; instead, we observe an algebraic decay proportional to $t^{-(k+1)}$ for all the values of k we have considered. Fourth, a dispersion in the angular momentum removes the divergence of $\lambda(\omega)$ at the edge of the band which is responsible for the oscillating modes with $k \leq 1$ in the reduced model, such that these modes disappear when the mass of the gas is small enough, and it suppresses the coherent response which is present when all the gas particles have the same angular momentum. **[(Erik) The last statement relies on the operator with dispersion and on the free solution; the PIC runs with dispersion of section VII are still to be done.]**

Particle methods have a long tradition in the numerical study of spherically symmetric collisionless systems, both in the Newtonian case [37, 38] and in the relativistic one, where they have been used to analyze the collapse of star clusters and the critical behavior at the threshold of black hole formation [12–14, 41, 42]; see Refs. [43–45] for introductions to the PIC method, Ref. [46] for a higher-order implementation in the context of plasma physics, and Ref. [47] for a convergence proof of a PIC scheme for the spherically symmetric Vlasov-Einstein system. Measuring the response of the gas to a small perturbation with such a method is delicate, because the signal has to be separated from the noise which is due to the finite number of computational particles. In this article we address this problem by placing the particles on a regular lattice in the action-angle variables of the self-consistent steady state, by encoding the perturbation in the weights of the particles instead of their positions (as in Refs. [48, 49]), and by subtracting a reference run without perturbation. We also show that the observables need to be computed with the action-angle variables belonging to the total potential, since those of the external potential alone leave a time-independent residue which could be mistaken for an undamped mode.

The remainder of this article is organized as follows. In section II we formulate the Vlasov-Poisson system with an external potential in spherical symmetry, for the reduced model with fixed angular momentum as well as for a distribution of angular momenta, and we introduce the action-angle variables and the observables used throughout this article. In section III we review the mixing phenomenon in the absence of self-gravity, derive the linearized equations about a steady state and introduce the gain $\lambda(\omega)$ of the self-consistent loop. Our numerical methods are described in section IV, and their validation is presented in section V. The results for the reduced model are discussed in section VI, and those for a distribution of angular momenta in section VII. Conclusions and an outlook to future work are given in section VIII. Technical details regarding the numerical computation of the action-angle variables are included in Appendix A.

II. THE VLASOV-POISSON SYSTEM IN SPHERICAL SYMMETRY

In this section we formulate the Vlasov-Poisson system with an external potential and reduce it to spherical symmetry, first for a gas whose particles have arbitrary angular momenta and then for the reduced model in which all of them have the same magnitude of the angular momentum. Next, we introduce the dimensionless variables used in our numerical computations, the action-angle variables associated with a time-independent potential and, finally, the observables on which our analysis of the mixing phenomenon is based.

We consider a collisionless gas consisting of identical particles which are subject to a time-independent external potential Φ_{ext} and to the gravitational potential Φ_{gas} generated by the gas itself. Since all the gas particles have the same mass, it is convenient to normalize the DF such that $f(t, x, p)d^3x d^3p$ is the mass of the gas contained in the volume element $d^3x d^3p$ of phase space, where from now on p denotes the momentum per unit mass.² With these conventions, the Vlasov-Poisson system consists of the Vlasov equation

$$\frac{\partial f}{\partial t} + p \cdot \nabla_x f - (\nabla_x \Phi) \cdot \nabla_p f = 0, \quad \Phi := \Phi_{ext} + \Phi_{gas}, \quad (1)$$

and the Poisson equation

$$\Delta \Phi_{gas} = 4\pi G \rho, \quad \rho(t, x) := \int_{\mathbb{R}^3} f(t, x, p) d^3p, \quad (2)$$

² This is equivalent to working in units for which the mass of each gas particle is one, as in Ref. [18].

where G denotes Newton's constant and ρ is the mass density of the gas. To obtain a unique solution of Eq. (2) we require that $\Phi_{gas}(t, x) \rightarrow 0$ for $|x| \rightarrow \infty$.

For the external potential we choose H  non's isochrone potential [1, 50, 51]

$$\Phi_{ext}(x) = -\frac{GM}{b + \sqrt{b^2 + r^2}}, \quad r := |x|, \quad (3)$$

which is generated by a spherical matter distribution of total mass M . The parameter $b \geq 0$ is a length scale which describes the transition from a harmonic potential (when $r \ll b$) to the potential of a point particle of mass M (when $r \gg b$). For $b = 0$, Eq. (3) reduces to the Kepler potential $\Phi_{ext} = -GM/r$, which is the external potential considered in Ref. [34]. The reason for this choice is that in this potential the period of the radial motion depends only on the energy of the particle [50, 51], and that the corresponding action-angle variables are known in closed form [1]. When the self-gravity of the gas is neglected, this provides exact solutions against which our numerical methods can be validated.

A. Reduction to spherical symmetry

A spherically symmetric gas configuration is described by a DF which is invariant with respect to simultaneous rotations of x and p . Such a DF depends on (x, p) only through the radius r , the radial component $p_r := x \cdot p/r$ of the momentum and the magnitude $L := |x \wedge p|$ of the angular momentum, that is,

$$f(t, x, p) = \mathcal{F}(t, r, p_r, L). \quad (4)$$

In this case, the density ρ and the potential Φ_{gas} depend only on t and r , and the Vlasov equation (1) reduces to

$$\frac{\partial \mathcal{F}}{\partial t} + p_r \frac{\partial \mathcal{F}}{\partial r} - \frac{\partial V_L}{\partial r} \frac{\partial \mathcal{F}}{\partial p_r} = 0, \quad V_L(t, r) := \Phi(t, r) + \frac{L^2}{2r^2}, \quad (5)$$

with V_L the effective potential for the radial motion. Since no derivative with respect to L appears in Eq. (5), the gas particles with a given value of L behave like a one-dimensional gas in the potential V_L . However, particles with different values of L interact with each other through the potential Φ_{gas} , which is determined by the spherically reduced Poisson equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi_{gas}}{\partial r} \right) = 4\pi G \rho, \quad \rho(t, r) = \frac{2\pi}{r^2} \int_0^\infty \int_{-\infty}^\infty \mathcal{F}(t, r, p_r, L) dp_r L dL, \quad (6)$$

see for instance Ref. [21]. The solution of Eq. (6) which is regular at the center and vanishes at infinity is

$$\Phi_{gas}(t, r) = -4\pi G \int_0^\infty \frac{\rho(t, s)}{\max\{r, s\}} s^2 ds, \quad (7)$$

and its gradient is $\partial \Phi_{gas} / \partial r = Gm(t, r)/r^2$, where $m(t, r) := 4\pi \int_0^r \rho(t, s) s^2 ds$ is the mass of the gas contained in the sphere of radius r at time t .

For the following, it is useful to note that when a spherically symmetric function is integrated over phase space, the volume element can be replaced with

$$d^3x d^3p = 8\pi^2 dr dp_r L dL. \quad (8)$$

In particular, the total mass and the total energy of the gas are given by

$$M_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) dr dp_r L dL, \quad (9)$$

$$\mathcal{E}_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) \left[\frac{p_r^2}{2} + \frac{L^2}{2r^2} + \Phi_{ext}(r) + \frac{1}{2} \Phi_{gas}(t, r) \right] dr dp_r L dL, \quad (10)$$

both of which are conserved in time. The conservation of $\mathcal{E}_{gas}$ provides a test for the accuracy of our numerical evolutions, see section V.

B. The reduced model with fixed angular momentum

A further simplification, which has been considered in Refs. [18, 21, 34], is obtained by assuming that all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum.³ In this case, the DF is of the form

$$\mathcal{F}(t, r, p_r, L) = \mathcal{F}_0(t, r, p_r) \delta(L - L_0), \quad (11)$$

and Eqs. (5,6) yield

$$\frac{\partial \mathcal{F}_0}{\partial t} + p_r \frac{\partial \mathcal{F}_0}{\partial r} - \frac{\partial V_{L_0}}{\partial r} \frac{\partial \mathcal{F}_0}{\partial p_r} = 0, \quad \rho(t, r) = \frac{2\pi L_0}{r^2} \int_{-\infty}^{\infty} \mathcal{F}_0(t, r, p_r) dp_r, \quad (12)$$

where the potential Φ_{gas} is still given by Eq. (7). Likewise, Eqs. (9,10) hold with $\mathcal{F}$ replaced by $\mathcal{F}_0$, L by L_0 , and the integral over L by the factor L_0 . Equations (12,7) constitute a Vlasov-Poisson system on a two-dimensional phase space with coordinates (r, p_r) . It describes a gas of concentric spherical shells which move in the effective potential V_{L_0} . The centrifugal term in this potential prevents the shells from reaching the center, and the gas configurations considered in this article are confined to a region of the form $0 < r_{in} \leq r \leq r_{out}$. In spite of its simplicity, the reduced model retains the two ingredients which are relevant for this article, namely the dependency of the frequency of the radial motion on the orbit, which is responsible for the mixing, and the self-gravity of the gas.

In the second situation considered in this article, $\mathcal{F}$ is a regular function of L whose support is contained in an interval $[L_1, L_2]$ with $0 < L_1 < L_2$. In this case, the phase space is three-dimensional, and the orbits with different values of L have different frequencies.

C. Dimensionless variables

In our numerical computations we use dimensionless variables, which are obtained by measuring masses in units of M , lengths in units of b and times in units of the dynamical time

$$t_{dyn} := \sqrt{\frac{b^3}{GM}}. \quad (13)$$

Accordingly, p_r is measured in units of $\sqrt{GM/b}$, L in units of $\sqrt{GMb}$, the potentials and the energy per unit mass in units of GM/b , and $\mathcal{F}$ in units of $M/(GMb)^{3/2}$. In terms of these variables, all the equations of this section retain their form with $G = M = b = 1$; in particular, the isochrone potential is $\Phi_{ext}(r) = -1/(1 + \sqrt{1 + r^2})$. The resulting system contains no free parameters, and the strength of the self-gravity is controlled by the ratio M_{gas}/M between the mass of the gas and the mass which generates the external potential. In the limit $M_{gas}/M \rightarrow 0$ the potential Φ_{gas} can be neglected, and one recovers the model analyzed in Ref. [18].

In the Kepler case $b = 0$ the external potential does not provide a length scale. However, in this case the system (5,6) is invariant with respect to the rescaling

$$(t, r, p_r, L, \mathcal{F}, \Phi) \mapsto (\alpha^{3/2}t, \alpha r, \alpha^{-1/2}p_r, \alpha^{1/2}L, \alpha^{-3/2}\mathcal{F}, \alpha^{-1}\Phi), \quad \alpha > 0, \quad (14)$$

which leaves M_{gas} unchanged. Therefore, b can be replaced with an arbitrary length in the definitions above, such that $\Phi_{ext}(r) = -1/r$, and in the reduced model this freedom can be used to fix the value of L_0 .

To give an idea of the physical time scale, we note that $M = 10^5 M_\odot$ and $b = 1$ pc, which are of the order of magnitude of the mass and of the core radius of a globular cluster [1], yield $t_{dyn} \approx 4.7 \times 10^4$ yr. As will be seen below, $2\pi t_{dyn}$ is a lower bound for the periods of the radial motion in the isochrone potential.

D. Steady states and action-angle variables

Suppose that the total potential Φ is independent of time. Then, the energy per unit mass

$$E := \frac{p_r^2}{2} + V_L(r) \quad (15)$$

³ Note that in Ref. [34] the symbol L refers to the square of the magnitude of the angular momentum.

is conserved along the trajectories of the gas particles, and it follows that any DF of the form

$$\mathcal{F}(r, p_r, L) = \psi(E, L), \quad (16)$$

with a nonnegative function ψ , is a time-independent solution of the Vlasov equation (5). In order for Eq. (16) to provide a solution of the Vlasov-Poisson system, the potential Φ_{gas} , which enters the definition of E through V_L , has to coincide with the one generated by the density of the DF (16) through Eqs. (6,7). This is a nonlinear problem for Φ_{gas} , whose numerical solution is described in section IV. In the following, a solution $(\mathcal{F}_{eq}, \Phi_{eq})$ of this problem, with Φ_{eq} the total potential and ρ_{eq} the corresponding density, will be referred to as a self-consistent steady state.

Next, we introduce action-angle variables for the radial motion in a time-independent potential Φ , following Refs. [1, 18]. In our applications, Φ is either the external potential (3) or the total potential Φ_{eq} of a self-consistent steady state. In both cases $\Phi'(r) = Gm_{tot}(r)/r^2$, where $m_{tot}(r)$ denotes the total mass contained in the sphere of radius r , including the one which generates the external potential. Since the function $r \mapsto rm_{tot}(r)$ increases strictly monotonically from 0 to ∞ , for each $L > 0$ the effective potential V_L has a unique critical point $r = r_{min}(L)$. It is determined by the equation $r^3\Phi'(r) = L^2$, it is a non-degenerate minimum of V_L , and it corresponds to a circular orbit with energy $E_{min}(L) := V_L(r_{min}(L))$. Since $V_L(r) \rightarrow \infty$ for $r \rightarrow 0$ and $V_L(r) \rightarrow 0$ for $r \rightarrow \infty$, the orbits with energies in the range $E_{min}(L) < E < 0$ are bound, and they oscillate between the turning points $r_1(E, L) < r_2(E, L)$, which are the two solutions of the equation $V_L(r) = E$.

As in Ref. [18] we define the area function

$$A(E, L) := \oint p_r dr = 2 \int_{r_1(E, L)}^{r_2(E, L)} \sqrt{2(E - V_L(r))} dr, \quad (17)$$

which represents the area enclosed by the orbit in the (r, p_r) -plane, and whose partial derivative with respect to E yields the period of the radial motion,

$$T(E, L) := \frac{\partial A}{\partial E}(E, L) = 2 \int_{r_1(E, L)}^{r_2(E, L)} \frac{dr}{\sqrt{2(E - V_L(r))}}. \quad (18)$$

The action and angle variables associated with the radial motion are

$$J := \frac{A(E, L)}{2\pi}, \quad Q := \frac{2\pi}{T(E, L)} \int_{\gamma_r} \frac{dr}{p_r}, \quad (19)$$

where γ_r denotes the segment of the orbit in the (r, p_r) -plane which connects, in the clockwise direction, the inner turning point $(r_1(E, L), 0)$ with the given point (r, p_r) . Hence, Q/Ω with $\Omega := 2\pi/T(E, L)$ is the time which has elapsed since the last passage of the particle through its pericenter; in particular, $Q = 0$ at the pericenter, $Q = \pi$ at the apocenter, and $0 < Q < \pi$ when $p_r > 0$. For each fixed value of L , the map $(r, p_r) \mapsto (Q, J)$ is symplectic, that is $dr \wedge dp_r = dQ \wedge dJ$, and it provides smooth coordinates on the set of bound orbits which are not circular. Since $T > 0$, the first relation in Eq. (19) can be inverted for each L , which yields the energy as a function $E(J, L)$ of the action variable and the angular momentum, and the frequency of the radial motion is

$$\Omega(J, L) = \frac{\partial E}{\partial J}(J, L) = \frac{2\pi}{T(E, L)}. \quad (20)$$

[(Erik) We denote the orbital frequency by Ω (it is called ω in Ref. [18]) and reserve ω for the frequency of the collective oscillation.] In terms of the variables (Q, J, L) , the volume element (8) is $8\pi^2 dQ dJ L dL$, and the Vlasov equation (5) for a time-independent potential assumes the simple form

$$\frac{\partial \mathcal{F}}{\partial t} + \Omega(J, L) \frac{\partial \mathcal{F}}{\partial Q} = 0. \quad (21)$$

In particular, the steady states (16) are precisely those DFs which are independent of the angle variable, and they can be regarded as functions of (J, L) .

For the external potential (3) the area function can be computed explicitly, see Refs. [1, 18]. In terms of the dimensionless variables introduced in the previous subsection, one finds

$$A(E, L) = 2\pi \left[\frac{1}{\sqrt{-2E}} - c(L) \right], \quad E_{min}(L) = -\frac{1}{2c(L)^2}, \quad (22)$$

with $c(L) := (L + \sqrt{L^2 + 4})/2$ for the isochrone potential and $c(L) := L$ for the Kepler potential. This yields

$$E(J, L) = -\frac{1}{2[J + c(L)]^2}, \quad \Omega(J, L) = \frac{1}{[J + c(L)]^3} = (-2E)^{3/2}. \quad (23)$$

Therefore, the frequency depends only on the energy, it decreases monotonically when J or L increase, and for each fixed L its maximum is attained in the limit $J \rightarrow 0$ of circular orbits. For the isochrone potential, $c(L) > 1$ implies that $\Omega < 1$, such that the periods of the radial motion are larger than 2π . An explicit expression for the angle variable Q in terms of (r, p_r, L) can be found in section 3.5.2 of Ref. [1]; its inversion requires the solution of an equation of Kepler's type. In contrast to this, for a self-consistent steady state the potential Φ_{eq} is only known numerically, and in this case we compute the action-angle variables by numerical quadrature of Eqs. (17,18,19), see Appendix A.

E. Observables

As in Ref. [18], we describe the macroscopic state of the gas through observables of the form

$$N_\varphi(t) := \int f(t, x, p) \varphi(x, p) d^3x d^3p, \quad (24)$$

where φ is a given test function on phase space. In this article we consider two classes of such observables.

The first one refers to the action-angle variables (Q, J) belonging to a time-independent reference potential. Unless otherwise stated, this is the total potential Φ_{eq} of the self-consistent steady state whose perturbations are analyzed, which reduces to Φ_{ext} when the self-gravity of the gas is neglected. For a spherically symmetric configuration and a test function of the form $\varphi = a(Q)B(J, L)$, with real-valued functions a and B , Eqs. (8,24) and the expansion of a in a Fourier series yield

$$N_\varphi(t) = \sum_{k \in \mathbb{Z}} \hat{a}_k^* h_k(t), \quad \hat{a}_k := \frac{1}{2\pi} \int_0^{2\pi} a(Q) e^{-ikQ} dQ, \quad (25)$$

with the functions

$$h_k(t) := 8\pi^2 \int_0^\infty \int_0^\infty \int_0^{2\pi} \mathcal{F}(t, Q, J, L) B(J, L) e^{-ikQ} dQ dJ L dL, \quad k \in \mathbb{Z}. \quad (26)$$

In the reduced model, B depends only on J and Eq. (26) holds with $\mathcal{F}$ replaced by $\mathcal{F}_0$ and the integral over L by the factor L_0 . Since $\mathcal{F}$ is real, $h_{-k} = h_k^*$, such that it is sufficient to consider $k \geq 0$. For a steady state h_k vanishes for all $k \neq 0$, whereas h_0 is equal to the mass of the gas weighted with the function B . Therefore, the functions $h_k(t)$ with $k \neq 0$ measure the departure of the gas configuration from a steady state, and their decay in time is the signature of the relaxation process we are interested in. We stress that this property relies on the use of the action-angle variables which belong to the total potential Φ_{eq} . If those of the external potential are used instead, a self-consistent steady state gives rise to nonvanishing, time-independent values of h_k with $k \neq 0$, as we show in section V.

The second class of observables consists of the perturbation of the gravitational potential,

$$\delta\Phi(t, r) := \Phi(t, r) - \Phi_{eq}(r) = -4\pi G \int_0^\infty \frac{\rho(t, s) - \rho_{eq}(s)}{\max\{r, s\}} s^2 ds, \quad (27)$$

where we have used Eq. (7). For each fixed r this is the difference between the values of the observable (24) with test function $\varphi(x, p) = -G/\max\{r, |x|\}$ which belong to the DF of the gas and to the one of the steady state. In contrast to the functions $h_k(t)$, the quantity $\delta\Phi$ does not refer to any particular coordinates on phase space, and together with its gradient it is the one whose decay is addressed in Refs. [34, 35]. To measure its magnitude we use the root mean square

$$\|\delta\Phi(t)\| := \left(\frac{1}{r_b - r_a} \int_{r_a}^{r_b} \delta\Phi(t, r)^2 dr \right)^{1/2} \quad (28)$$

over an interval $[r_a, r_b]$ which covers the region occupied by the gas.

III. MIXING AND LINEAR RESPONSE

[(Erik) To be written: mixing without self-gravity, linearized equations, gain of the self-consistent loop, the case with a distribution of angular momenta.]

IV. NUMERICAL METHODS

[(Erik) To be written: particle-in-cell method, equilibria and initial data, linearized solver, evaluation of the gain.]

V. CODE VALIDATION

[(Erik) To be written.]

VI. RESULTS FOR FIXED ANGULAR MOMENTUM

[(Erik) To be written.]

VII. RESULTS FOR A DISTRIBUTION OF ANGULAR MOMENTA

[(Erik) To be written.]

VIII. CONCLUSIONS

[(Erik) To be written.]

Acknowledgments

[(Erik) To be written.]

Appendix A: Numerical computation of the action-angle variables

[(Erik) To be written.]

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